Classification Decision Trees

Data Mining
Lecture # 12

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Measuring Node Impurity

- p(i|t): fraction of records associated with node t belonging to class i

$$\text{Entropy}(t) = -\sum_{i=1}^{c} p(i\,|\,t)\log p(i\,|\,t)$$

- Used in ID3 and C4.5

$$\text{Gini}(t) = 1 - \sum_{i=1}^{c} [p(i\,|\,t)]^2$$

- Used in CART, SLIQ, SPRINT.

$$\text{Classification error}(t) = 1 - \max_i [p(i\,|\,t)]$$

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Example

C1	**0**
C2	**6**

P(C1) = 0/6 = 0   P(C2) = 6/6 = 1
Gini = 1 – (P(C1))^2 – (P(C2))^2 = 1 – 0 – 1 = 0
Entropy = – 0 log 0 – 1 log 1 = – 0 – 0 = 0
Error = 1 – max (0, 1) = 1 – 1 = 0

C1	**1**
C2	**5**

P(C1) = 1/6   P(C2) = 5/6
Gini = 1 – (1/6)^2 – (5/6)^2 = 0.278
Entropy = – (1/6) $\log_2$ (1/6) – (5/6) $\log_2$ (1/6) = 0.65
Error = 1 – max (1/6, 5/6) = 1 – 5/6 = 1/6

C1	**2**
C2	**4**

P(C1) = 2/6   P(C2) = 4/6
Gini = 1 – (2/6)^2 – (4/6)^2 = 0.444
Entropy = – (2/6) $\log_2$ (2/6) – (4/6) $\log_2$ (4/6) = 0.92
Error = 1 – max (2/6, 4/6) = 1 – 4/6 = 1/3

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Impurity measures

- All of the impurity measures take value zero (minimum) for the case of a pure node where a single value has probability 1
- All of the impurity measures take maximum value when the class distribution in a node is uniform.

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Splitting Based on GINI

- When a node p is split into k partitions (children), the quality of split is computed as,

$$GINI_{split} = \sum_{i=1}^{k} \frac{n_i}{n} GINI(i)$$

where, n_i = number of records at child i,
n = number of records at node p.

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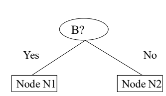
Binary Attributes: Computing GINI Index

- Splits into two partitions
- Effect of Weighing partitions:
  - Larger and Purer Partitions are sought for.



	Parent
C1	**6**
C2	**6**
Gini = 0.500	

Gini(N1)
= 1 – (5/7)^2 – (2/7)^2
= 0.194

Gini(N2)
= 1 – (1/5)^2 – (4/5)^2
= 0.528

	N1	N2
C1	**5**	**1**
C2	**2**	**4**
Gini=0.333		

Gini(Children)
= 7/12 * 0.194 +
5/12 * 0.528
= 0.333

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Categorical Attributes

- For binary values split in two
- For multivalued attributes, for each distinct value, gather counts for each class in the dataset
  - Use the count matrix to make decisions

Multi-way split

	CarType		
	Family	Sports	Luxury
C1	1	2	1
C2	4	1	1
Gini	0.393		

Two-way split
(find best partition of values)

	CarType	
	{Sports, Luxury}	{Family}
C1	3	1
C2	2	4
Gini	0.400	

	CarType	
	{Sports}	{Family, Luxury}
C1	2	2
C2	1	5
Gini	0.419	

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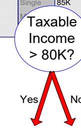
Continuous Attributes

- Use Binary Decisions based on one value
- Choices for the splitting value
  - Number of possible splitting values = Number of distinct values
- Each splitting value has a count matrix associated with it
  - Class counts in each of the partitions, A < v and A ≥ v
- Exhaustive method to choose best v
  - For each v, scan the database to gather count matrix and compute the impurity index
  - Computationally Inefficient! Repetition of work.

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No			No
10	No			Yes



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Continuous Attributes

- For efficient computation: for each attribute,
  - Sort the attribute on values
  - Linearly scan these values, each time updating the count matrix and computing impurity
  - Choose the split position that has the least impurity

Sorted Values

Cheat	No	No	No	Yes	Yes	Yes	No	No	No	No
Taxable Income	60	70	75	85	90	95	100	120	125	220

Split Positions

	55		65		72		80		87		92		97		110		122		172		230	
	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>
Yes	0	3	0	3	0	3	0	3	1	2	2	1	3	0	3	0	3	0	3	0	3	0
No	0	7	1	6	2	5	3	4	3	4	3	4	3	4	4	3	5	2	6	1	7	0
Gini	0.420		0.400		0.375		0.343		0.417		0.400		***0.300***		0.343		0.375		0.400		0.420	

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Splitting based on impurity

- Impurity measures favor attributes with large number of values
- A test condition with large number of outcomes may not be desirable
  - # of records in each partition is too small to make predictions

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Splitting based on INFO

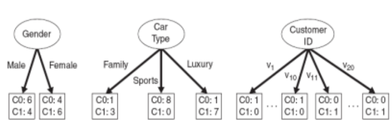

Figure 4.12. Multiway versus binary splits.

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Gain Ratio

- The information gain measure tends to prefer attributes with large numbers of possible values i.e. biased towards tests with many outcomes.
- Gain ratio: a modification of the information gain that reduces its bias on high-branch attributes.
- Gian ratio should be
  - Large when data is evenly spread
  - Small when all data belong to one branch
- Gain ratio takes number and size of branches into account when choosing an attribute
- It corrects the information gain by taking the intrinsic information of a split into account

- $SplitInfo_A(D) = -\sum_{j=1}^{v} \frac{|D_j|}{|D|} \times \log_2\left(\frac{|D_j|}{|D|}\right).$

$GainRatio(A) = \frac{Gain(A)}{SplitInfo_A(D)}.$

$\boxed{Split(S,A) \equiv -\sum \frac{|S_i|}{|S|} \log_2 \frac{|S_i|}{|S|}.}$

$GainRatio(S,A) = \frac{Gain(S,A)}{Split(S,A)}$

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Gain Ratio

- Adjusts Information Gain by the entropy of the partitioning (SplitINFO). Higher entropy partitioning (large number of small partitions) is penalized!
- Used in C4.5
- Designed to overcome the disadvantage of impurity

Example (Buy Computers) :

$$SplitInfo_{income}(D) = -\frac{4}{14}\log_2\left(\frac{4}{14}\right) - \frac{6}{14}\log_2\left(\frac{6}{14}\right) - \frac{4}{14}\log_2\left(\frac{4}{14}\right) = 1.557$$

$$Gain(income) = 0.029 \text{ bits}$$

$$GainRatio(income) = \frac{Gain(income)}{SplitInfo(income)} = 0.019$$

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Example (Play tennis) :

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$Split(S,Day) = 14 \times \left(-\frac{1}{14}\log\frac{1}{14}\right) = 3.807$$

$$Gain\ Ratio(S,Day) = \frac{0.94}{3.807} = 0.25$$

$$Split(S,Outlook) = \left(-\frac{5}{14}\log\frac{5}{14}\right) \times 2 + \left(-\frac{4}{14}\log\frac{4}{14}\right) = 1.577$$

$$Gain\ Ratio(S,Outlook) = \frac{0.247}{1.577} = 0.157$$

$$Gain\ Ratio(S,Humidity) = \frac{0.152}{1} = 0.152$$

$$Gain\ Ratio(S,Temperature) = \frac{0.029}{1.362} = 0.021$$

$$Gain\ Ratio(S,Wind) = \frac{0.048}{0.985} = 0.049$$

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More on the gain ratio

- "Outlook" still comes out top
- However: "ID code" has greater gain ratio
  - Standard fix: In particular applications we can use an *ad hoc* test to prevent splitting on that type of attribute
- Problem with gain ratio: it may overcompensate
  - May choose an attribute just because its intrinsic information is very low
  - Standard fix:
    - First, only consider attributes with greater than average information gain
    - Then, compare them on gain ratio

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Comparing Attribute Selection Measures

- The three measures, in general, return good results but
- Information Gain
  - Biased towards multivalued attributes
- Gain Ratio
  - Tends to prefer unbalanced splits in which one partition is much small than the other
- Gini Index
  - Biased towards multivalued attributes
  - Has difficulties when the number of classes is large
  - Tends to favor tests that result in equal-sized partitions and purity in both partitions

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Stopping Criteria for Tree Induction

- Stop expanding a node when all the records belong to the same class
- Stop expanding a node when all the records have similar attribute values

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Decision Tree Based Classification

- Advantages:
  - Inexpensive to construct
  - Extremely fast at classifying unknown records
  - Easy to interpret for small-sized trees
  - Accuracy is comparable to other classification techniques for many simple data sets

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Example: C4.5

- Simple depth-first construction.
- Uses Information Gain
- Sorts Continuous Attributes at each node.
- Needs entire data to fit in memory.
- Unsuitable for Large Datasets.
  - Needs out-of-core sorting.

- You can download the software from:
  http://www.cse.unsw.edu.au/~quinlan/c4.5r8.tar.gz

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